

DIVYAM PANDEY

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EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science & Engineering (Data Science)

• **CGPA: 9.09 / 10**

• **Relevant Coursework:** Machine Learning, Artificial Intelligence, Information Retrieval, Natural Language Processing, Probability & Statistics, Data Structures & Algorithms, Database Management Systems

Aug. 2023 – Present

Vellore, Tamil Nadu

EXPERIENCE

Ernst & Young Global Delivery Service (EY GDS)

June 2026 – Aug 2026

Intern

Kochi, Kerala

• Evaluated retrieval quality and ranking behavior of production LLM-based search pipelines across distributed enterprise architectures, following training in Deep Learning, NLP, Graph RAG, and Advanced RAG architectures.

Mahindra & Mahindra

June 2025 – July 2025

AI Intern

Mumbai, Maharashtra

- Improved retrieval relevance in enterprise chatbots by engineering a Retrieval-Augmented Generation (RAG) pipeline using LangChain and vector embeddings.
- Enabled complex, multi-step reasoning and cross-session memory by designing stateful AI workflows utilizing LangGraph.
- Improved enterprise workflow efficiency by automating manual processes using n8n and integrating the Model Context Protocol (MCP) for secure tool orchestration.

PROJECTS

Vidhan-AI (Legal RAG Assistant with Learning-to-Rank)

Mar 2026 – Jul 2026

LangChain, ChromaDB, BGE-1.5, PyTorch, LightGBM

- github.com/El3troX/vidhan_ai – Built a Retrieval-Augmented Generation assistant answering natural-language queries over Indian statutory law (Constitution, BNS, BNSS), indexing 200+ documents into 1,390 structure-aware, citation-preserving chunks using ChromaDB and BGE-large-1.5 embeddings.
- Engineered a learning-to-rank re-ranking layer (PyTorch RankNet, LightGBM LambdaRank) combining dense embedding similarity, BM25, and exact-citation-match features atop the retrieval pipeline.
- Improved ranking quality over the embedding-only baseline by 17.6% relative Precision@3 and 9.6% relative MRR, and lifted baseline nDCG@5 by 18.2% absolute (0.70 → 0.83) by fixing ingestion/chunking issues; selected LightGBM over PyTorch for sub-millisecond CPU production inference.

Valkyrie-AML (Agentic Anti-Money-Laundering Platform)

2026

Next.js, FastAPI, LangGraph, scikit-learn, SHAP, NetworkX

- github.com/El3troX/Valkyrie-AML – Engineered an agentic AML investigation platform combining Random Forest/Isolation Forest detection, SHAP explainability, and NetworkX-based Personalized PageRank risk propagation, achieving 82.7% F1 (73.0% precision, 95.3% recall) across 200,000 transactions.
- Built an 8-tool LangGraph-orchestrated agent that routes natural-language compliance queries to detection, fund-flow-tracing, and SAR-generation tools, automating FinCEN-style Suspicious Activity Report drafting via LLM with a deterministic fallback.

Market Pulse

May 2026

R, ARIMA, SARIMA, ETS, GARCH

- github.com/El3troX/marketpulse – Recorded ~12% MAPE on 30-day out-of-sample forecasts by developing a time-series forecasting and model evaluation framework.
- Validated predictive accuracy across 5+ years of NIFTY-50 data (~1,250 sessions) using 6-fold time-series cross-validation.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), SQL, R

ML & IR: PyTorch, LightGBM, scikit-learn, Random Forest, Isolation Forest, SHAP, Learning-to-Rank (RankNet, LambdaRank), NDCG/MRR/Precision@k, RAG Systems, LangChain, LangGraph

Search & Retrieval Infra: ChromaDB, Vector Embeddings, Sentence-Transformers (MiniLM, BGE-1.5), NetworkX

Databases: MySQL, PostgreSQL

Data & Tools: NumPy, Pandas, Matplotlib, Git, GitHub, VS Code, n8n, Streamlit, Next.js, FastAPI

CERTIFICATIONS, HACKATHONS & LEADERSHIP

Certifications: Google Data Analyst Certificate, Google (2026) · ML Specialization, Coursera – Andrew Ng (2025) · MERN Stack Development, Ethnus (2025)

Hackathons: IIT Bombay Eureka 25 · Deloitte Hacksplasion · HackWithInfy · Goldman Sachs India Hackathon (2025–2026)

Leadership: Senior Core Member, Technology and Gaming Club, VIT (2023–Present) – coordinated cross-team logistics and scheduling for Riviera and Gravitas events, serving 300+ participants.